Assignment L-4-c: 2026: Artifact Selection

PI (1.3)

Now that you have generated a rich set of possible artifacts, the important project constraints (**musts**), and team desires (**wants**) the team must now select and obtain its artifact.

The Assignment (You will use the summary sheet you constructed with the previous assignment)

Your specific assignment is as follows:

1. Create a **KTDA** table (or use the supplied one) using the artifacts, constraints, and desires (you named) from assignment L-4-b. It is required now to add "**instructor approval**" constraint to your constraints if it was not among your selected set of constraints.
   Note: if, after you apply the constraints, there are fewer than **three** feasible[1] artifacts left, go back to your prioritized list of artifacts and find some additional feasible artifacts, or relax the constrains (subject of instructor approval) so that you apply your desires (**wants**) to at least three feasible artifacts.
2. For the top ***three*** ideas in the KTDA table, develop the **Adverse Consequences Table**.
3. Based on, the combination of the evaluation out of the KTDA table and the assessment of the adverse consequences, **exclude** one of the artifacts. Make sure you still have ***at least two*** artifact names available for comparison.
4. Prepare a detailed **cost estimate** for the remained ***two artifacts***. This cost estimate should include costs of time, materials, tools and any other additional expenditure needed for implementation.
5. Based on the results, in the KTDA, Adverse Consequences Tables and the cost analysis, **select** the team's artifact.
6. It is not recommended to have the same artifact for more than one team in the same group. If your team selected an artifact which was selected by another team before; you need to go to another choice. This must be reported in the assignment scenario.

The Work in the Notebook

- Again, as you do this work, develop the work using the presentation sandwich. We (the Laboratory Instructors, your customers) are **NOT LOOKING** for a report; rather we are looking in your Design Notebook for an organized presentation of your work. **The work need not be word-processed** but it must be neat and readable. Word processing everything is not a way to exceed expectations. It is expected that at least part of all your discussions will be a reflection back on the process that was undertaken in doing the completed task.

- Much of this work will be done as a team during a team meeting. It is important to not lose the discussion that takes place as the team creates the various tables and finally arrives at its final decision. Don't forget to include evidence that you are functioning as a team (meeting agendas, minutes, etc.)

- **For those who may be confused between the assessment section and the project section**. The **assessment** section must have the original task checklist (i.e the 1st, 2nd …etc.) in addition to the resubmission checklist. The **project section,** however, must have the task itself (L-4-a, L-4-b covered by the corresponding **PTW** checklist, no cover sheets.

The Assessment

- The work will be self-assessed according to the 3rd DNB assessment checklist items. Peer assessment will be done in class under the supervision of the instructor. The instructor will do the final assessment.
- Upon the final assessment of this task, the Third Notebook Checklist and the resubmission checklist (if used) must be placed in the assessment section. The associated project assignment, however, covered by the PTW checklist shall be in the project section.

[1] Feasible artifacts are artifacts that satisfy all the constraints (**musts**)

Third Notebook Assessment Checklist (2026) (PI:1.3)

Team_____ Work Products Being Evaluated: Project: (Assignment #L-4-c)

1. Work Evaluated by (self?)_____ Assessment Symbol/Color _____ Date_____
2. Work Evaluated by (peer?)_____ Assessment Symbol/Color _____ Date_____
3. Work Evaluated by (instructor)_____ Assessment Symbol/Color _____ Date_____

This Checklist must be completed (self-assessed) before submission for assessment

Yes	No	Self-Regulation Issues: (*half lapse will be issued (to each member) for each NO*)
		A. This Checklist is **completed** (self-assessed) before submission & is located in the assessment section
		B. There is a miscellaneous section that contains what is not enough to open separate section; If there is no such section type 'NA' in the yes cell (miscellaneous section comes after the discussion section)
		C. The DNB is solid, strong, cohesive, in good shape and all included papers are placed with the correct orientation
		D. There are no other Self-Regulation problems (if No state the point here ……)
		Enter the total number of issued lapses

Yes	No	Expected Features	Re
		1. The work (L-4-c) meets expectations for the PTW. (*PTW checklist is the cover sheet of L-4-c*)	
		2. The **TOC** has been updated to reflect the new work. (*Page numbers are shown for **all tasks***)	
		3. There is a completed **KTDA** Table with at least **THREE** feasible (satisfies all **musts**) ideas.	
		4. **Every** weight/rate/number in the **KTDA** is defined and its way of calculation is explained	
		5. There is a completed **Adverse Consequences Table** for at least three (artifacts).	
		6. Every probability/seriousness in the **Consequences** Table is defined and explained	
		7. In general; there are **NO** errors in the calculations required for each table.	
		8. There is a detailed **cost estimate** for the **implementation process** of the top **two** artifacts	
		9. There is an **explicit** statement -at least in concluding the work- defining the team's selected artifact.	
		10. There is at least one meeting minute for the meeting(s) devoted to finish this work	
		11. The work used in creating "**team process check**" and its checklists are in the assignment section	
		12. The work used in creating "**team norms**" and its checklists are in the assignment section	
		13. The **L-3-discussion** is updated and located by the end of the DNB (before any miscellaneous)	
		A specific cell in the most right column is to be checked in the re-submission only	

Wow	Ok	Weak	Revealed Features
			1. Rate the **quality of the DNB organization** (*e.g., use of unique page numbers, reference to other work in the notebook, sense of continuous work on a single large project consisting of numerous tasks, etc.*)
			2. Rate the **quality of the context** (of assignment L-4-c)
			3. Rate the **readability** of the L-4-c (*i.e., the level of communication "**noise**," which consists of grammatical, spelling, syntax, etc., errors that hinder communication*)
			4. Rate the **quality** of the effort shown in generating the **KTDA** & the **Adverse Consequences** Tables (*e.g., amount and type of discussion used in filling in the two tables, the rationale in defining the relative weights of the desires, i.e., **wants**, the rationale for the selection of the idea*)
			5. Rate the **quality of the Discussion** (of assignment L-4-C) (e.g., *how good was the reflection?*)
			6. Rate the **quality of team meeting material** (e.g., explicit roles, agendas, minutes, team processing, etc.)

Yes	Exciting Feature(s)

summary			Modified the instructor has the right to cancel one wow in-front of a "No" or a "weak"		
Wow+ Exciting Features	Weak	NO	Wow+ Exciting Features	Weak	NO

Results of Initial Assessment (based on the modified part of the previous table)

E	**All** Yes for everything, **no** Weak , at least two WOWs or One WOW and an Exciting Feature, (Absolutely no lapses)
M	**All** Yes for the Expected Features (only), **one**-Weak is accepted
NI	**Either (one** No for Expected Features) **OR** (**two** Weaks) …… (it is OR not and)
NCE	**Any other combination**
NS	The folder is not rigid enough to be handled by the grader or was not submitted

For instructor use only: *check the box to the right if it was required (from this team) to upload the word-processed part of this assignment to the BB for the percentage of the plagiarism. The upload must be before C-5-3.* ☐

*If the work was assessed as **NI, NCE or NS** you may, following the information given in the LAB-FDM and the instructions given on the Resubmission of Work Checklist (if applicable), improve the work and resubmit it (if applicable) for re-assessment.*

Team:	Date: / /202	***KT Decision Analysis: Ses: L-5-2 – (starting 472)***											
Alternative:		**A :**		**B :**		**C :**		**D :**		**E :**		**F :**	
Musts	Instructor approval	GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO	
		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO	
		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO	
		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO	
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		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO		GO NO GO	

Wants	Weight	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score
1.													
2.													
3.													
4.													
5.													
6.													
		Total A =		***Total B =***		***Total C =***		***Total D =***		***Total E =***		***Total F =***	

Assignment L-4-d: *IMPLEMENTATION* 2026

Please refer to course books for the implementation section

PI (3.2)

Now that you have selected the **team's artifact** your team needs to obtain the artifact, dissect it and prepare for the final class. The assembly may be approved by a **W**illing **W**orkers; (**WW** could be an expert technician, a college professor …etc.).

The Assignment: Your specific assignment is detailed as follows:

1. Perform a Situation Analysis (**KTSA**) to prioritize the tasks and activities remain and required to be completed for this project. Don't forget to include the documentation and project management tasks you are expected to do as you work on this project. At least four tasks/situations are to be discussed.

2. Based on results of your KTSA; develop a schedule (a **Gantt Chart**) that shows when the tasks (for this "d" assignment) will be completed. The duration of each task is to be estimated based on your experience in doing similar tasks. The schedule should indicate who is going to do each task.

3. Construct the network diagram of all steps (activities) required for this "d" assignment. Identify the **Critical Path** to determine the total required activities to be done and completed. This network should follow the way you worked (or will work) on performing the assignment of "food for thought".

4. If you have not already done so, obtain your team's artifact (in L-4-b it is stated that: *There are, of course, other ways besides buying a new item to obtain a desired artifact: used items, or something donated*).

5. **In C-6-2 (and out of C-6-2)**; you are expected to get the artifact and an appropriate set of tools to GE211-class and do the following (see next page):-

a) Isometric-free-hand-sketch: make free hand sketch for the artifact (as one piece) in three dimensional view. Sketch must be completely freehand and according to the engineering drawing rules. It is required to point and name the essential sides of the artifact (base – front – top …etc). [*Isometric: drawing representing three-dimensional objects with equal axes*].

b) Dissection[1] (disassembly): four sub-steps are required (Plan-Dissect-Create-Report)

(b-i) The team needs to PLAN and document the expected steps for dissecting the artifact. This will be called the **I**nitial **D**isassembly **P**lan **IDP**. The IDP *may be subject* to WW's approval or modifications (if available). It is expected that your first attempts at the disassembly plan will not be your last. (*a sample of the IDP is given by the end of this document*).

(b-ii) DISSECTION: now you need to take your artifact apart to its lower level (using your IDP). Stick firmly to your IDP & the comments from the WW (if any). Facing problems or obstacles, try to modify the steps of the IDP or get back to your WW (out of class) to make suggestions. Do not forget to record the new modifications to the IDP. The modified (revised) dissection plan will be called FDP (**F**inal **D**isassembly **P**lan). After finishing, append the FDP by the name of the set of tools you used in this dissection. You may discuss with your fellows the sort of the expected problems and how to go around from the beginning. (*Treat the FDP as if given to anyone, he can complete the dissection without getting back to you*).

(b-iii) Upon artifact Dissection: CREATE free-hand-sketches (drawings) of all the parts. Be sure to give each part a name and part number. It should be clear on the sketches (drawings) who did what. Sketches must be completely freehand and according to the engineering drawing rules. At least two different projections are to be sketched for each part.

(b-iv) Prepare a TECHNICAL REPORT for your artifact that describes "How Stuff Works"- *a sample is given by the end of this document*.

c) Re-assemble the artifact: now it is the time to reassemble the artifact.

- The first logic plan is to reverse the order of the FDP and replace some instructions by the opposite of it (*as take off by place, turn LHS by turn RHS …etc.*). The obtained plan now will be called the **I**nitial **A**ssembly **P**lan **IAP**. It is expected that your first attempts at an assembly plan will not be your last.
- Consider the same tools you recommended in the **FDP** for assembly according to IAP.
- Brainstorm (*potential assembly problems, their consequences, what would be their possible causes, how to prevent these from happing and what to do if happened*). These will be your elements of PPA task (this task is necessary in L-4-e and no need to report its results here).
- Based on the previous steps, you may start assembling the artifact.
- Upon proceeding with the assembly: always stick to the steps of IAP & the brainstormed considerations. If you stocked at certain point, or some steps need to be modified or checked by the WW, record that for future use in the "e" assignment.

Note that: at least one contribution is needed from the WW for any of the (IDP, FDP, IAP)

[1] The question of how far the dissection should be taken always comes up. There is no correct answer for this question, but the team should bear in mind the following two constraints (based on which, the team must decide how far to take the dissection): (1) The WW may asses the dissection plan and (2) the instructor will have no screwdrivers and any other wrenches required for hand work.

The Work in The Notebook

- Again (how many times is this going to appear?), as you do your work completing this assignment, present the work using the presentation sandwich. We (the Laboratory Instructors, your customers) are **NOT LOOKING** for a report; rather we are looking in your Design Notebook for an organized presentation of your work. **The work need not be word-processed** but it must be neat and readable. Word processing everything is not a way to exceed expectations. It is expected that at least part of all your discussions will be a reflection back on the process that was undertaken in doing the completed task. You may find it helpful to put the sketches together in a separate tabbed section of the notebook (with a sandwich structure). Much of this work will be done as a team during a team meeting.

- For those who are still confused between the assessment section and the project section: See similar section in the L-4-c assignment.

The Assessment

The work will be self-assessed according to the 4th DNB assessment checklist items. Peer assessment will be done in class under the supervision of the instructor. The instructor will do the final assessment.

Sample for the IDP (consider that the artifact name is x) 1- Place x on its base 2- Prepare a small box to collect the little pieces 3- locate screw (number 3) 4- Use a Philips screwdriver (size ??) to take the screw (number 3) out. (turn LHS) 5- Place the screw (number 3) in the box you have 6- Repeat 3-5 fro screw drivers (# ?????) 7- Side number ??) is now free; carefully take it out 8- etc	**Sample for the technical report (consider that the artifact name is x)** 1- Place three-batteries size (AA) in the battery compartment number (xxx) 2- Place the artifact in right up position 3- Press button (yyy) or turn it on 4- etc

Fourth Design-Notebook Assessment Checklist – (2026) [PI:3.2]

*There is no **RESUBMISSION** of this work. The grade granted during the first assessment is final and is applied to the entire team members*

Team_____ Work Products Being Evaluated: Project: (Assignment #L-4-d)

1. Work Evaluated by (self?)_____ Assessment Symbol/Color _____ Date_____
2. Work Evaluated by (peer?)_____ Assessment Symbol/Color _____ Date_____
3. Work Evaluated by (instructor)_____ Assessment Symbol/Color _____ Date_____

This Checklist must be completed (self-assessed) before submission for assessment

Yes	No	**Self-Regulation Issues: (*half lapse will be issued (to each member) for each NO*)**
		A. This Checklist is **completed** (self-assessed) before submission & is located in the assessment section
		B. There is a miscellaneous section that contains what is not enough to open separate section; If there is no such section type 'NA' in the yes cell (miscellaneous section comes after the discussion section)
		C. The DNB is solid, strong, cohesive, in good shape and all included papers are placed with the correct orientation
		D. The DNB assessment summary sheet is completed
		E. There are no other Self-Regulation problems (if No state the point here ……)
		Enter the total number of issued lapses

Yes	No	**Expected Features (*most of the points are about L-4-d*)**
		1. **TOC** is updated **without** any little problem
		2. The work (L-4-d) meets expectations for the **PTW** which is *the cover sheet of L-4-d*
		3. The level of communication "noise" [*which consists of grammatical, spelling, syntax, etc. errors*] is less than 5
		4. There is a completed **KTSA** Table (with at least four tasks/situations)
		5. There is a discussion of how the results of the **KTSA**-Table was (were) reached
		6. There is a **Gantt Chart** that shows the timing of the tasks shown in the **KTSA** table
		7. There is a **Critical Path** schedule – for how to complete this assignment
		8. There is an **isometric hand sketch** (IHS)for the artifact
		9. There is an **IDP**
		10. There is a **FDP** with name of the tools required for dissection
		11. There are some **free-hand-sketches** (FHS) of the artifact components (at least one from each member)
		12. There is a technical report for the artifact that describes "How Stuff Works".
		13. There is an **IAP**
		14. There is at least one contribution from the **WW**
		15. There is at least one meeting minute for the meeting(s) devoted to finish this work
		16. The discussion of assignment **L-3** is updated and located by the end of the DNB

Wow	Ok	Weak	**Revealed Features (*most of the points are about L-4-d*)**
			1. Rate the quality of the **context** of L-4-d
			2. Rate the quality of the **effort** shown in generating and using the **KTSA** Table (e.g., amount and type of discussion used in completing the table)
			3. Rate the quality of the **IDP, FDP and IAP.**
			4. Rate the quality of IHS&FHS (paid effort(s) in creating them, amount, presentation, look, following the engineering drawing rules)
			5. Rate the **quality of Gantt Chart and schedule creation effort** (e.g., completeness in tasks and assignments, the relationship with the KTSA work)
			6. Rate the quality of the effort shown in generating the **Critical Path** schedule
			7. Rate the quality of the **technical report** for the artifact that describes "How Stuff Works".
			8. Rate the quality of **team meeting material** (e.g., explicit roles, agendas, meeting minutes, team processing, etc.)

Yes	**Exciting Features**

Results of FINAL Assessment (*The equivalent/modified assessment system is not to be applied in this assignment*)

E	At least two WOWs or (One WOW and an Exciting Feature), and Zero weak, Zero NO, Zero lapses
M	All Yes for the Expected Features – up to two weaks is permitted here
NI	No more than Four as the total sum of weaks and Nos
NCE	Any other combination
NS	The folder is not rigid enough to be handled by the grader or was not submitted

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